

CRYSTAL PM
HOW-TO GUIDE

Set up a Crystal Meet room

Put a Crystal Meet device in a conference room and join Zoom and Google Meet calls from the room's screen, step by step.

This guide takes one conference room from an empty Raspberry Pi to a room that shows up in the Crystal Meet dashboard and joins calls from its own web page. Plan about 45 minutes per room, most of it waiting for downloads. No programming experience is needed: if you can follow a recipe and type the commands printed here, you can do this. Repeat it for each room.

STEP 1

Prepare the device

Install Raspberry Pi OS, sign in, connect the TV and camera.

STEP 2

Create the room

Add the room on the dashboard's **Provisioning** page and copy its token.

STEP 3

Install Crystal Meet

One command with the room's config file.

STEP 4

First run and screens

Join a test call, then point the table screen and the door sign at the device.

BEFORE YOU BEGIN

Hardware per room. Raspberry Pi 4 (4 GB) or Pi 5, the official power supply, a 32 GB microSD card, an HDMI cable to the room's TV, and a USB camera and speakerphone (for example a Logitech Brio 505 and a Jabra Speak2 55).

Two screens on the network. A touch tablet or PoE touch panel on the table to join and control meetings, and a small screen by the door to show whether the room is free. Neither connects to the Pi: each only needs a browser, power, and the office network. Give the room device a fixed address on your router (a DHCP reservation) so the screens can always find it.

The dashboard address. The Crystal Meet dashboard must be reachable from the room's network, for example <http://192.168.1.20:3000> for the dashboard and port **3001** for devices. If the dashboard runs on a Windows PC under WSL2, turn on mirrored networking or forward ports 3000 and 3001 first, or the devices will show Offline.

Accounts. A dashboard sign-in with the admin role, and the Zoom or Google Meet links you want to test with.

1 Prepare the device

- **Flash the card.** On your computer, open [Raspberry Pi Imager](#), choose your Pi model, then [Raspberry Pi OS \(64-bit\)](#) with desktop, and your microSD card.
- **Pre-fill the settings.** When Imager offers customisation, set the hostname to something like [crystal-meet-room-1](#), the username to [pi](#) with a password you keep, your Wi-Fi network and your time zone, and turn on SSH. Write the card.
- **Connect and boot.** Put the card in the Pi, plug in the TV over HDMI, the camera and speakerphone over USB, then power. Wait for the desktop to appear on the TV.
- **Turn on automatic sign-in.** Open [Preferences > Raspberry Pi Configuration > System](#) and set [Auto Login](#) to on, so the room comes back by itself after a power cut.

You should now see the Raspberry Pi desktop on the room's TV, and the Pi should be on the office network.

2 Create the room in the dashboard

- **Open the dashboard** at its address on port 3000 and sign in.
- **Go to [Provisioning](#)** and fill in the [Room Name](#) exactly as people call the room (for example [Room 1](#)) and, if you like, its [Location](#). Press [Generate Token](#).
- **Copy the token** shown under [Enrollment Token](#). It is a long string of letters and numbers; you will paste it into the device's config in the next step.

You should now see the token on the page and, on [Devices](#), the room listed as provisioning until its device connects.

GOOD TO KNOW

The token links one device to this room and works once. If a device is ever reinstalled, create a new token for it. Nothing else about the room needs to change.

3 Install Crystal Meet on the device

- Open a terminal on the Pi (the black icon in the top bar), or connect over SSH from your computer.
- Get the software. Type `sudo apt update && sudo apt install -y git`, then `git clone https://github.com/ben-abeo/croom.to.git`, then `cd croom.to`.
- Make the room's config. Type `cp deploy/rooms/room-1.yaml ~/room.yaml` (use `room-2.yaml` or `room-3.yaml` for the other rooms), then `nano ~/room.yaml`.
- Fill in three values. The room's **name** and **location**; the dashboard **url** as `http://<dashboard address>:3001`; and the **enrollment_token** you copied. Save with **Ctrl+O**, Enter, then **Ctrl+X**.
- Run the installer. Type `sudo bash installer/install.sh --config ~/room.yaml`. It installs the browser and the room software; expect 10 to 20 minutes.
- Start it. When the installer prints **Crystal Meet installation complete**, type `sudo systemctl start croom`. The service also starts on every boot from now on.

You should now see a browser window open on the TV. That window is the room's meeting screen; leave it open.

4 First run and the screens

- Open the room page. On your laptop or phone on the office network, go to `http://crystal-meet-room-1.local:8080`, or use the device's IP address with `:8080`.
- Check the dashboard. On **Devices**, the room should show **online** within a minute.
- Join a test call. Paste a Zoom link, or a Zoom meeting ID, into **Join with a link** and press **Join**. The TV shows the meeting joining; the page turns blue and offers **Mute**, **Turn camera off** and **Leave**.
- Leave. Press **Leave**, then **Tap again to leave**.
- Set up the table screen. On the tablet or panel, open its kiosk browser (Fully Kiosk Browser on Android, Guided Access with Safari on an iPad) and set the start page to `http://<device address>:8080/`. Turn on full screen and keep the screen awake.
- Set up the door sign. On the door screen, set the start page to `http://<device address>:8080/sign`. It shows green when the room is free, amber when a meeting is about to start, and red while the room is in use or booked.

You should now see the room page say the room is free again, the door sign green, and the dashboard still showing the device online.

TIP

Zoom links that include their passcode (the part after `?pwd=`) join straight through. Google Meet joins as a guest, so someone in the meeting admits the room when it asks.

Troubleshooting

The room page does not load

On the Pi, type `sudo systemctl status croom`. If it is not running, type

`sudo journalctl -u croom -n 50` and look for the first line marked ERROR. Make sure port 8080 is not used by something else on the device.

The device shows offline

From the Pi, type

`curl http://<dashboard address>:3001/health`.

If nothing comes back, the dashboard is not reachable from the room's network: check the address in `~/room.yaml` and the dashboard PC's networking. If the token was already used, create a new one and run the installer again with the updated config.

The browser opens but never joins

Zoom needs the passcode inside the link; paste the full invitation link. Google Meet waits until someone in the meeting admits "the room". If the page says it could not join, press **Dismiss** and try the link again.

The screens cannot open the page

Use the device's IP address rather than its `.local` name; many tablets do not resolve those.

Check that the screen and the device are on the same network, then open

`http://<device address>:8080/api/status` from a laptop to confirm the device answers.

No browser window opens on the TV

The service waits for the desktop before it starts, so give a freshly booted Pi a minute. If the window still does not appear, type

`sudo journalctl -u croom -n 50` and look for "Failed to initialize". The browser is installed under `/opt/croom/browsers`; if that folder is missing, run the installer again.

No sound or picture in the call

Check the USB camera and speakerphone are connected before the device starts, then reboot the Pi. The service runs as the signed-in desktop user so it can use the room's screen and audio.

GOOD TO KNOW

Under the hood the device software is still called `croom`: that is the name you will see in service and log commands. Everything people see says Crystal Meet.